Solved Midterm Paper	CS607- Artificial Intelligence	
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MIDTERM EXAMINATION
Spring 2011
CS607- Artificial Intelligence

Question No: 1 (Marks: 1) - Please choose one
Breadth-first search is a good idea when you are confident that the branching factor is_____

►Extremely small
►**Small**
►Medium
►Large

Question No: 2 (Marks: 1) - Please choose one
Another expert system named_____was developed by Digital Equipment Corporation, as a computer configuration assistant.

►**R1/XCON (Page 112)**
►MYCIN
►Dendral
►R3/XCON

Question No: 3 (Marks: 1) - Please choose one
An AI system must form a meaningful and useful _____ of the internal information.

►**Representation (Page 89)**
►Execution
►Learning
►Planning

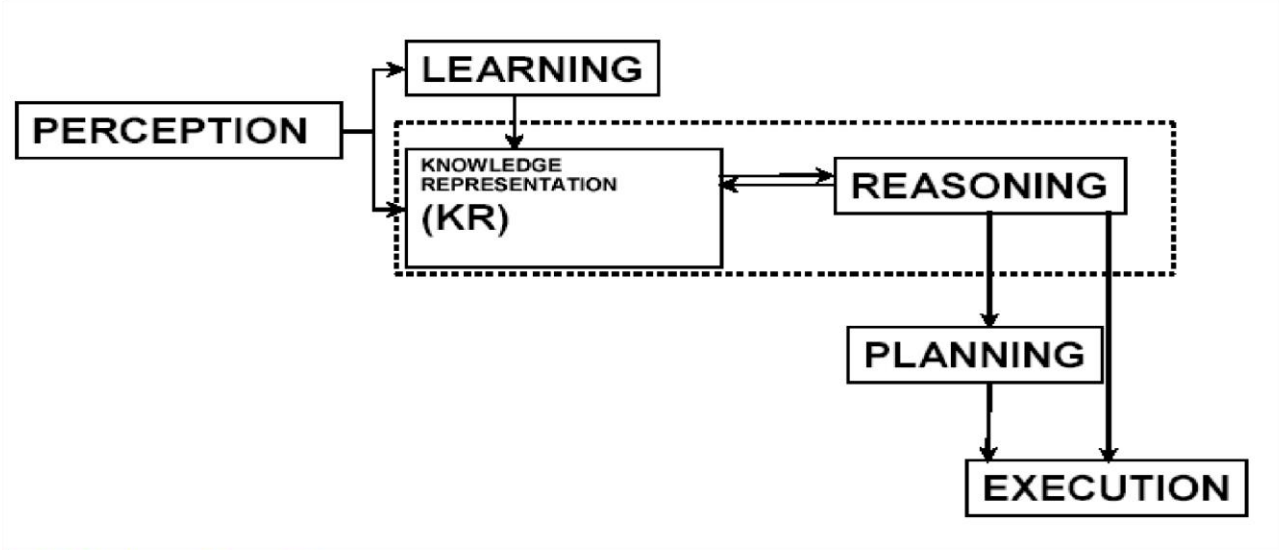
Question No: 4 (Marks: 1) - Please choose one
An AI system has a _____ component that allows the system to get information from its environment.

►Planning
►**Perception (Page 89)**
►Learning
►Execution

Question No: 5 (Marks: 1) - Please choose one
Progressive deepening guarantees to find the solution at a minimum depth like
►DFS
►**BFS (Page 37)**
►None

Question No: 6 (Marks: 1) - Please choose one
In Adversarial search the goals of the adversaries are usually _____ to each other
►**Contrary (Page 62)**
►Same
►None

Question No: 7 (Marks: 1) - Please choose one
Which Cycle does the following figure show?



►**AI Cycle (Page 89)**
►Design Cycle
►SDLC
►None

خدا کے سوا کسی سے امید مت رکھو

Question No: 8 **Please choose one**
To infer new information from semantic networks, we can ask questions from nodes.

►**True (Page 97)**
►False

Question No: 9 (Marks: 1) - Please choose one
Which one is not the application area of expert system?

►Diagnosis
►Prescription
►Interpretation
►**None (Page 114)**

Ref: - Expert systems may be used in a host of application areas including diagnosis, interpretation, prescription, design, planning, control, instruction, prediction and simulation.

Question No: 10 (Marks: 1) - Please choose one
An expert system is different from conventional programs in the sense that program control and knowledge are _____.

►**Separate (Page 121)**
►Defined
►Together
►Common

Question No: 11 (Marks: 1) - Please choose one
Which one of the following is involved in an ES development project:

►The domain expert
►The knowledge engineer
►The end user
►**All of the given (Page 122)**

Question No: 12 (Marks: 1) - Please choose one
Semantic networks are computationally expensive at _____

►**Run time (Page 97)**
►Compile Time
►Start Time
►End Time

MIDTERM EXAMINATION
Spring 2011
CS607- Artificial Intelligence

Question No: 1 (Marks: 1) - Please choose one
We can get optimal solution given some parameters using Genetic Algorithm.

►**True (Page 79)**
►False

Question No: 2 (Marks: 1) - Please choose one
_____ reasoning is based on forming, or inducing a „generalization" from a limited set of observations.

►Deductive
►Abductive
►Analogical
►**Inductive (Page 102)**

Question No: 3 (Marks: 1) - Please choose one
_____ is the process of deriving logical conclusions from given facts.
►Representation
►Execution
►**Reasoning (Page 102)**
►Planning

Question No: 4 (Marks: 1) - Please choose one
Identify the correct step used to start design of an expert system.
►**Feasibility study (Page 129)**
►Problem reorganization
►Scope study
►Rapid prototyping

Question No: 5 (Marks: 1) - Please choose one
Inductive learning is based on the knowledge that if something happens a lot it is likely to be generally _____
►**True (Page 160)**
►False

►Ambiguous
►None of the given

Question No: 6 Please choose one
If the antecedent is only partially true, then the output fuzzy set is truncated according to the _____ method

►Intrinsic
►Implication (Page 153)
►Boolean
►None of the given

Question No: 7 (Marks: 1) - Please choose one
Choose the fields in which Fuzzy inference systems have been successfully applied:

►automatic control
►data classification
►decision analysis
►All of the given (Page 153)

Question No: 8 (Marks: 1) - Please choose one
Usually a _____ graph is chosen to represent a fuzzy set.
►Triangular (Page 151)
►Circular
►Conical
►None of the given

Question No: 9 (Marks: 1) - Please choose one
Fuzzy logic is actually a superset of conventional Boolean logic
►TRUE (Page 150)
►FALSE

Question No: 10 (Marks: 1) - Please choose one
Reasoning in fuzzy logic is just a matter of generalizing the familiar _____ logic.

►Boolean (Page 147)
►Complex
►Coagnitive
►Supervised

Question No: 11 (Marks: 1) - Please choose one
A classical set is a container, which wholly includes or wholly excludes any given element.

►**TRUE** **(Page 145)**
►FALSE

Question No: 12 (Marks: 1) - Please choose one
The degree of truth that we have been talking about is specifically driven out by a function called the _____ function.

►**Membership (Page 149)**
►Ordinary
►Fuzzy
►Inline

Question No: 13 (Marks: 1) - Please choose one
The tractable problems are further divided into structured and _____ problems

►Non-structured
►**Complex (Page 166)**
►Simple

Question No: 14 (Marks: 1) - Please choose one
If the antecedent is only partially true, then the output fuzzy set is truncated according to the _____ method

►Intrinsic
►**Implication (Page 153) rep**
►MSE stands for
►Mean Square Error

MIDTERM EXAMINATION
Spring 2010
CS607- Artificial Intelligence (Session - 1)

Question No: 1 (Marks: 1) - Please choose one
"The branch of computer science that is concerned with the automation of intelligent behavior" this definition is from:

► **Luger and Stubblefield (Page 8)**
► Winston
► Schalkoff
► Bellman

Question No: 2 **Please choose one**
Searching is a formal mechanism to explore
►**Alternatives (Page 21)**
►Recursive
►Best
►Fitness

Question No: 3 (Marks: 1) - Please choose one
In Artificial Intelligence GA stands for Genetic Algorithms

► **True (Page 77)**
► False

Question No: 4 (Marks: 1) - Please choose one
---------- is based on forming, or inducing a „generalization" from a limited set of observations

► **Inductive reasoning (Page 102)**
►Deductive reasoning
► Analogical reasoning
► Common-sense reasoning

Question No: 5 (Marks: 1) - Please choose one
"A computer program designed to model the problem solving ability of a human expert" is known as ---

►**Expert system (Page 111)**
►Intelligent System
►Echo System
►Energy System

Question No: 6 (Marks: 1) - Please choose one
An expert system may replace the expert or assist the expert

► **True (Page 113)**
► False

Question No: 7 (Marks: 1) - Please choose one
A ---------- is „A person who posses the skill and knowledge to solve a specific problem in a manner superior to others"

►**The domain expert (Page 122)**
►The knowledge engineer

►The end user
►All of the given

Question No: 8 (Marks: 1) - Please choose one
Hill Climbing is basically a ---------- with a measure of quality that is assigned to each node in the tree.

►Depth First Search (Page 39)
►Breadth First Search
►Best First Search
►Beam Search

Question No: 9 (Marks: 1) - Please choose one
---------- are closely coupled components; each is intrinsically tied to the other.
i-Knowledge
representation ii-
Reasoning iii-Planning iv-
Execution
► ii & iii
► ii & iii
► iii & iv
► i & ii (Page 89)

Question No: 10 (Marks: 1) - Please choose one
Every graph can be converted into a tree
► True (Page 22)
► False

Question No: 11 (Marks: 1) - Please choose one
Which one of the problem is more subtle, and consequently, is more frustrating:

►Foothill problem
►Plateau
►Ridge (Artificial Intelligence, 3/E) click here for detail
►Box

MIDTERM EXAMINATION
Spring 2009
CS607- Artificial Intelligence (Session - 1)

Question No: 1 (Marks: 1) - Please choose one
Most of the solution spaces for problems can be represented in a _____

►**Graph (Page 21)**
►Table
►Demo

Question No: 2 (Marks: 1) - Please choose one
By getting grips on _____ that deal with searching techniques in graphs and trees, problem solving can be performed in an efficient manner.

►Pseudocode
►**Algorithms (Page 21)**
►Charts
►Graphs

Question No: 3 (Marks: 1) - Please choose one
Every graph can be converted into a tree.

►**True (Page 22)**
►False

Question No: 4 (Marks: 1) - Please choose one
In Breadth First Search the node with the largest value of height will be at the _____ priority to be picked.

►**Maximum (Page 28)**
►Minimum
►None of the given

Question No: 5 (Marks: 1) - Please choose one
Breadth-First Search checks all paths of a given length before moving on to any longer paths.

►**True** Click here for detail
►False

Question No: 6
Breadth-first search is a good idea when you are confident that the branching factor is _____
►Extremely small
►**Small** **(Artificial Intelligence, 3/E)** click here for detail
►Medium
►Large

Question No: 7 **(Marks: 1) - Please choose one**
The foothhill problem occurs whenever there are_____ peaks.

►High
►**Secondary** **(Artificial Intelligence, 3/E)** click here for detail
►Primary
►Deep

Question No: 8 **(Marks: 1) - Please choose one**
The Plateau problem comes up when there is a mostly flat area _____ the peaks.

►**Separating** **(Artificial Intelligence, 3/E)** click here for detail
►Joining
►Over
►None of the given

Question No: 9 **(Marks: 1) - Please choose one**
Which one of the problem is more subtle, and consequently, is more frustrating:
►Foothill problem
►Plateau
►**Ridge** **(Artificial Intelligence, 3/E)** click here for detail Rep
►Box

Question No: 10 **(Marks: 1) - Please choose one**
The paths found by best-first search are likely to be _____ than those found with other methods.

►None of the given
►**Shorter** **(Artificial Intelligence, 3/E)** click here for detail
►Longer

Question No: 11 (Marks: 1) - Please choose one
In Basic Genetic Algorithm the term mutation refers to a small random _____.
►Number
►Change (Page 77)
►Operator
►Operand

Question No: 12
Which of the following two components are closely coupled and each is intrinsically tied to the other. i. Knowledge representation ii. Reasoning iii. Execution
iv. Planning

►i & iii
►ii & iii
►iii & iv
►i & ii (Page 89) rep

Question No: 13 (Marks: 1) - Please choose one
Semantic networks are graphs, with nodes representing _____ and arcs representing _____ between objects.

►objects, relationships (Page 97)
►relationships, distance
►objects, distance
►distance, relationships

Question No: 14 (Marks: 1) - Please choose one
A proposition is the statement of a _____.
►Fact (Page 98)
►Equation
►Action
►Theorem

Question No: 15 (Marks: 1) - Please choose one
_____ reasoning is based on forming, or inducing a „generalization" from a limited set of observations. ►Deductive
►Abductive
►Inductive (Page 102) rep
►Analogical

Question No: 16 (Marks: 1) - Please choose one

An _____ is “A computer program designed to model the problem solving ability of a human expert."

►**Expert system (Page 111) rep**

►Intelligent System

►Echo System

►Energy Sys

Question No: 17
Another expert system named _____ was developed by Digital Equipment Corporation, as a computer configuration assistant.

►R1/XCON (Page 112)
►MYCIN
►Dendral
►R3/XCON

Question No: 18 (Marks: 1) - Please choose one
An expert system may replace the expert or assist the expert.
►True (Page 113) rep
►False

Question No: 19 (Marks: 1) - Please choose one
Conventional programming focuses on _____, while ES programming focuses on _____

►Solution, Problem (Page 122)
►Problem, Solution
►Problem, Expert
►Solution, Expert

Question No: 20 (Marks: 1) - Please choose one
In backward chaining terminology, the hypothesis to prove is called the _____.

►Proof
►Goal (Page 126)
►Plan
►None of the given

CS607 – Quizzes (Quiz No.1)

Question # 1 of 10 (Marks: 1) - Please choose one
Some essential components of problem solving are Problem statement, _____, solution space and Operators

►Complex State
►Initial State
►Intermediate State
►Goal State (Page 17)

Question # 2 of 10 (Marks: 1) - Please choose one
The traveling inside a solution space requires something called as _____
►Operands
►Inner solution
►Space solution
►Operators (Page 18)

Question # 3 of 10 (Marks: 1) - Please choose one
Genetic Algorithms is a search method in which multiple search paths are followed in _____
►Series
►Parallel (Page 77)
►None of the give
►Sequential

Question # 4 of 10 (Marks: 1) - Please choose one
Which of the following disciplines provides us with the theories of structure and meaning of language

►Linguistic (Page 9)
►Philosophy
►Biology
►Psychology

Question # 5 of 10 (Marks: 1) - Please choose one
In optimal path searches we try to find the _____ solution
►Least
►Worst
►Least but not worst
►Best (Page 24)

6 of 10 (Marks: 1)

In Depth First Search the node with the largest value of height will be at the _____ priority to be picked.

►Minimum
►**Maximum (Page 25)**
►Zero
►Both Minimum and maximum

Question # 7 of 10 (Marks: 1) - Please choose one

In adversarial search there may occur such a scenario where two opponents also called _____ are searching for goal.

►**Adversaries (Page 62)**
►Friend
►Players
►Intruders

Question # 8 of 10 (Marks: 1) - Please choose one

From discipline _____ we have information about the network structure of a human brain and all the theories on functionalities of different human organs.

►Mathematics
►**Biology (Page 9)**
►Computer Science
►Psychology

Question # 9 of 10 (Marks: 1) - Please choose one

Hit and trial is classical approach to solve the _____ problem easily

►**Trivial (Page 15)**
►Medium

Question # 10 of 10 (Marks: 1) - Please choose one

We use graph to represent problems and their solution spaces.

►False
►**True (Page 22)**

Question # 1 of 10 (Marks: 1) - Please choose one

In GA, the random process is repeated until an individual with required _____ level is found.
Select correct option:

►Higher
►Lower
►**Fitness** **(Page 86)**
►Logical

Question # 2
Mutation can be as simple as just flipping a bit at random or any number of bits
►**True (Page 79)**
►False

Question # 3 of 10 (Marks: 1) - Please choose one
Genetic algorithm uses evolutionary techniques, based on function optimization and artificial intelligence, to develop a solution.
►**True** click here for detail
►False

Question # 4 of 10 (Marks: 1) - Please choose one
An AI system has a _____ component that allows the system to get information from its environment.
►Planning
►**Perception (Page 89)**
►Learning
►Execution

Question # 5 of 10 (Marks: 1) - Please choose one
In the worst case of semantic network, we may need to traverse the entire network and then discover that the requested info _____. ►**Does not exist (Page 97)**
►Exists
►Is incorrect
►Is correct

Question # 6 of 10 (Marks: 1) - Please choose one
A statement in conjunctive normal form (CNF) consists of _____
►**ANDs of Ors. (Page 107)**
►ANDs
►Ors
►Ors of ANDs

Question # 7 of 10 (Marks: 1) - Please choose one
What is the correct order for solving a problem using GA
I. Choose the best individuals from the population for crossover
II. Choose initial population
III. Evaluate the fitness of each individual

►I,II,III
►I,III,II

►II,I,III
►**II,III,I (Page 84)**

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Intelligence id the characteristics of:

►**Living things (Page 4)**
►All things
►None of them
►All of them

Question # 9 of 10 (Marks: 1) - Please choose one
Intelligence is the ability to

►**Think / learn/Plan/ Schedule (Page 5)**
►Recognize / Remember
►Problem Solving
►All of the Above

Question # 10 of 10 (Marks: 1) - Please choose one
Ability to tackle ambiguous and fuzzy problems demonstrate
►**Intelligence (Page 6)**
► Non intelligence behavior
►All of the given
►None of the given

Question # 1 of 10 (Marks: 1) - Please choose one
Can we precisely define Artificial Intelligence?
►Yes We Can
►**No we cannot (Page 14)**

Question # 2 of 10 (Marks: 1) - Please choose one
The traveling inside solution space requires something called

►Problem Statement
►Operands
►**Operators (Page 18)**
►Solution Space

Question # 3 of 10 (Marks: 1) - Please choose one

In DFS node with the largest value of height will be at Priority

►**Maximum (Page 25)**
►Minimum
►Zero
►Least

4

Try to catch out own thoughts as they go by is

►**Introspection (Page 28)**
►Psychology
►Both of above
►None of the above

Question # 5 of 10 (Marks: 1) - Please choose one

Classical way of problem solving

►GA
►**Generate and Test (Page 15)**

Question # 6 of 10 (Marks: 1) - Please choose one

A function by which we can tell which board position is nearer to our goal is called

►Alternative function
►Recursive function
►Best function
►**Fitness function (Page 83)**

Question # 7 of 10 (Marks: 1) - Please choose one

Procedures that search the solution space in an uninformed manner are usually costly with respect to _____.

►Time
►Space
►Time and space both
►**All of the given (Page 37)**

Question # 8 of 10 (Marks: 1) - Please choose one

Best first search is a greedy approach.

►**True (Page 47)**
►False

Question # 9 of 10 (Marks: 1) - Please choose one

Answering the Sequence Problem need

►**Intelligence (Page 15)**

►Ability to make plan

► Ability to schedule

► None of the given

Cs607 – Quizzes (Quiz No.2)

Question # 1 of 10 (Marks: 1) - Please choose one

IF A THEN B This can be considered to have a similar logical meaning as the following:

►**A -> B (Page 99)**
►A <-> B
►A <- B
►None of the given

Question # 2 of 10 (Marks: 1) - Please choose one

Within an expert system, the _____ contains facts about a specific subject area and rules that express the reasoning procedures of an expert on the subject.

► Inference engine
►Knowledge engineer
► **Knowledge base**
►None of the given options

Question # 3 of 10 (Marks: 1) - Please choose one

In the statement "IF A THEN B", B is called

►Antecedent
►**Consequent (Page 95)**

Question # 4 of 10 (Marks: 1) - Please choose one

In some cases, the rules provide more definite actions such as "move left" or "close door", in which case the rules are being used to represent _____.

► Recommendations
► **Directives (Artificial Intelligence, 3/E)** Click here for detail
►Relations
► None of the given options

Question # 5 of 10 (Marks: 1) - Please choose one Expert system can be expressed as:

► It provides tools for the management, delivery, tracking, and assessment of various types of employee learning and training
►The set of business processes, culture, and behavior required to obtain value from investments in information systems
►Used for finding the optimal solution for a specific problem by examining a very large number or possible solutions for that problem
►Intelligent technique for capturing tacit knowledge in a very specific and limited domain of human expertise, this knowledge is converted to rules that can be used throughout the entire organization click here for detail

Question # 6 of 10 (Marks: 1) - Please choose one
In general, the antecedent of a rule compares an object with a possible value, using an operator. **►True** click here for detail
►False

Question # 7 of 10 (Marks: 1) - Please choose one
IF temperature is below 0 THEN weather is cold The above rule is used to represent _____
►Recommendations
►Directives
►Relations (Page 96)
►None of the given options

Question # 8 of 10 (Marks: 1) - Please choose one
Which of the following is a valid example which represents a suitable antecedent in a rule?

►IF x > 3
►IF name is "Bob"
►IF weather is cold
► All of the given options

Question # 9 of 10 (Marks: 1) - Please choose one

From discipline of _____ we have the tools and techniques to investigate the human mind and ways to represent the resulting theories

►Computer Science
►Biology
► Mathematics
► **Psychology (Page 9)**

Question # 10 of 10 (Marks: 1) - Please choose one
Intelligence is the characteristic of

► **Living being (Page 4)**
►All things
►None of them
►All of them

Question # 1 of 10 (Marks: 1) - Please choose one
_____ AI actually tries to recreate the functions of the inside of the brain as opposed to simply emulating behavior

►Weak
► **Strong (Page 8)**
►Weak and Strong
►None of the given

Question # 2 of 10 (Marks: 1) - Please choose one
_____ AI treats the brain as a black box and just emulates its functionality.

► **Weak (Page 8)**
►Strong
►Weak and Strong
►None of the given